Article

Multi-heterojunctioned plastics with high thermoelectric figure of merit

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Conjugated polymers promise inherently flexible and low-cost thermoelectrics for powering the Internet of Things from waste heat[1,2]. Their valuable applications, however, have been hitherto hindered by the low dimensionless figure of merit (ZT)[3–6]. Here we report high-ZT thermoelectric plastics, which were achieved by creating a polymeric multi-heterojunction with periodic dual-heterojunction features, where each period is composed of two polymers with a sub-ten-nanometre layered heterojunction structure and an interpenetrating bulk-heterojunction interface. This geometry produces significantly enhanced interfacial phonon-like scattering while maintaining efficient charge transport. We observed a significant suppression of thermal conductivity by over 60 per cent and an enhanced power factor when compared with individual polymers, resulting in a ZT of up to 1.28 at 368 kelvin. This polymeric thermoelectric performance surpasses that of commercial thermoelectric materials and existing flexible thermoelectric candidates. Importantly, we demonstrated the compatibility of the polymeric multi-heterojunction structure with solution coating techniques for satisfying the demand for large-area plastic thermoelectrics, which paves the way for polymeric multi-heterojunctions towards cost-effective wearable thermoelectric technologies.

Harvesting energy from distributed heat is crucial for powering wearable electronics and the ubiquitous Internet of Things[1]. Flexible thermoelectric generators (TEGs) combine the merits of excellent attachability, no moving parts and high reliability, providing a direct route to convert waste heat into electricity, particularly on the human body and heat resources with complicated curvatures. Polymers, with their lightweight nature, abundant molecular availability, solution processability and inherent softness with low Young's modulus, are ideal candidates for wearable and conformable TEGs. However, their practical potential is limited owing to the low dimensionless figure of merit, ZT. Despite attempts by the scientific community to break this low-performance limitation through fine-tailoring conjugated backbones[3,4], designing functional side chains[6,7], manipulating the condensed structure[1,5,8] and engineering doping levels[9,10], the yielding ZT ranges from only 0.01 to 0.5, much lower than those of commercially available bulk materials (ZT_{298K} = 0.8–1.0)[11,12] and striking flexible inorganic candidates (ZT = 0.6–1.1)[13,14]. More importantly, the lack of a significant breakthrough in the ZT has dampened expectations after a decade of enthusiasm for thermoelectric plastics.

In the framework of the phonon-glass electron-crystal model, the ideal thermoelectric materials should address the conflicting need in electrical conductivity (σ) and thermal conductivity (κ), that is, approaching the crystal limit for charge transport while reaching the amorphous limit for phonon scattering[6,15,16]. Currently, by decoupling electron and phonon transport through multiscale microstructure engineering[17–19], many inorganic superlattices[20–22] and bulk crystals with two-dimensional layered structures[23–25] have satisfied the critical demands and induced striking ZT values by suppressing lattice thermal conductivity (κ_L) with enhanced interfacial phonon scattering. However, these robust methods are infeasible for polymers, owing to the lack of both an ordered crystal lattice in their crystalline/amorphous films and underlayer polymer re-dissolution during sequence solution coating (solvent corrosion) to impede the construction of ideal periodical structures. Under these circumstances, a suggested way to revolutionize plastic thermoelectrics is to devise a polymer-based hierarchical heterojunction nanostructure that follows a superlattice framework in periodical geometry but has a sandwiched bulk-heterojunction interface created by fine-tuned

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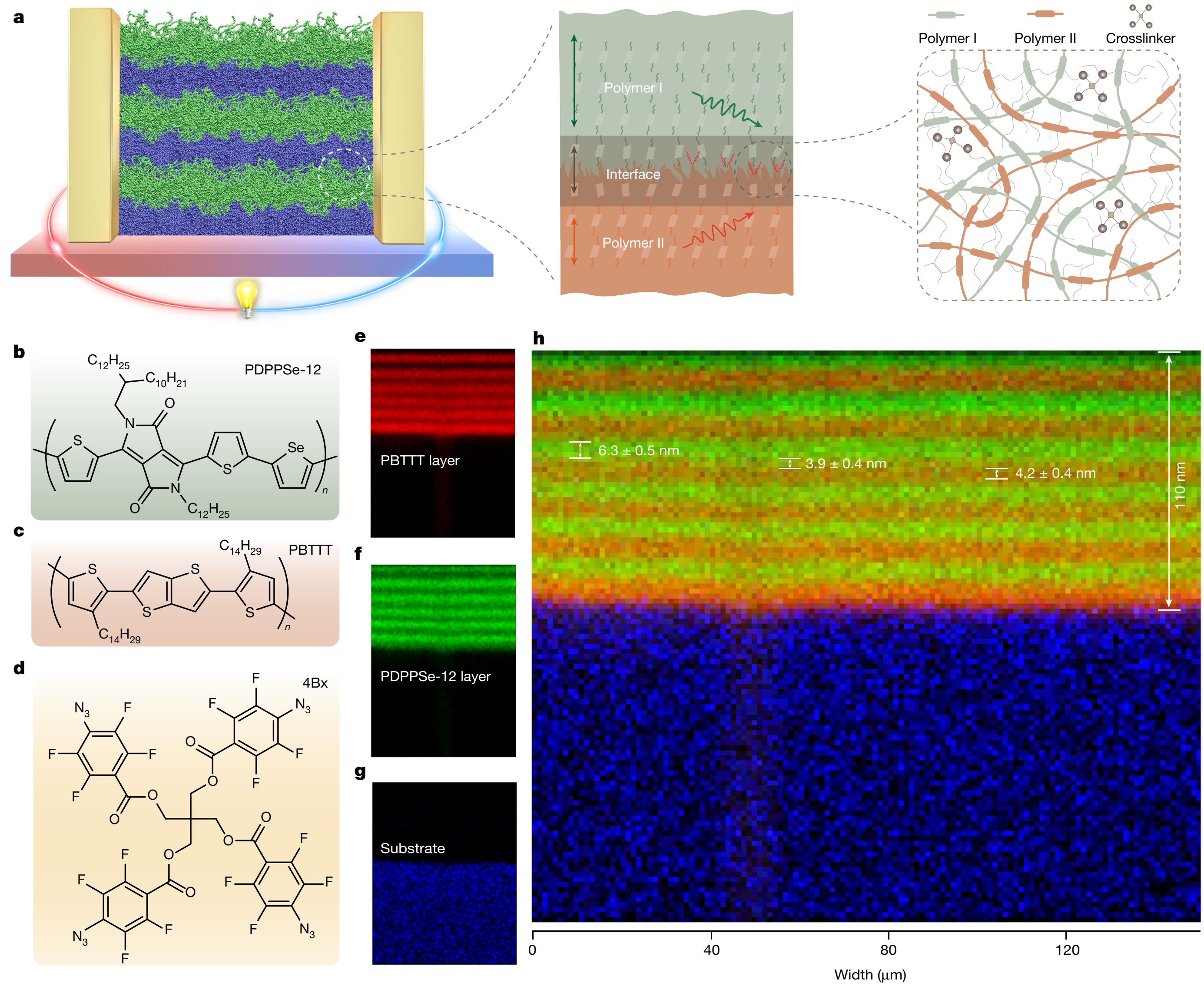

Fig. 1 | Concept and TOF-SIMS image of the PMHJ structure. a, Illustration of the PDPPSe-12:PBTTT PMHJ film. The PDPPSe-12 and PBTTT layers are depicted in blue and green, respectively. The colon in PDPPSe-12:PBTTT stands for a sequentially processed periodic film. PDPPSe-12 and PBTTT were photo-crosslinked by 4Bx. **b**–**d**, Chemical structures of PDPPSe-12 (**b**), PBTTT (**c**) and the crosslinker 4Bx (**d**). **e**–**g**, TOF-SIMS cross-sectional images along *y*–*z* axis of a single PDPPSe-12:PBTTT PMHJ (six period) film. The characteristic signals of the PDPPSe-12 layer (**e**; red), the PBTTT layer (**f**; green) and the substrate (**g**; blue) were extracted and reconstructed by software. **h**, The overlay of TOF-SIMS cross-sectional images of PDPPSe-12 (**e**), PBTTT (**f**) and the substrate (**g**). The periodic thicknesses are calibrated by step profiler and STEM-EDS measurements.

solvent corrosion. This unique nanostructure can fundamentally minimize κ, thus holding the potential to trigger a milestone ZT over 1.0 to rival commercial materials, which, however, has never been reported.

In this work, we propose a polymeric multi-heterojunction (PMHJ) with periodic dual-heterojunction features, where each period is composed of two distinct polymer layers and their sandwiched interpenetrating interfaces. By manipulating the individual polymer and interface thickness to sub-10 nm and sub-5 nm, respectively, the PMHJ film not only preserves a prominent power factor but also produces a low in-plane thermal conductivity ($\kappa_{\parallel}$) by enhancing interfacial propagon scattering. These eventually produce a maximum ZT of 1.28 at 368 K. The performance surpasses a value of 1.0 and outperforms commercially available thermoelectric materials in the near-room-temperature region. More importantly, the PMHJ structure is compatible with large-area solution coating techniques, making it a promising option for low-cost wearable TEGs.

PMHJ construction and characterization

High-mobility polymers are promising for overcoming the trade-off relation between the Seebeck coefficient (S) and σ. In contrast to conventional amorphous polymers, they typically show polycrystalline features, allowing for efficient charge transport and approaching an 'electron crystal' state. Meanwhile, their relatively stronger π–π interactions with ordered molecular packing, compared with the weak van-der-Waals-dominant interactions in amorphous polymers, cause them to deviate from the ideal 'phonon glass'. Instead, propagons, which behave as phonon-like propagating waves, may have a contribution in their thermal conductivity[26,27] (Supplementary Discussion I). We propose that there is a significant underestimation of this deviation in conventional belief, which leads to the disregard of the potential for maximizing thermoelectric performance through suppression of κ_L. This motivates us to define the PMHJ structure, which consists of a periodic arrangement of two polycrystalline polymer layers with

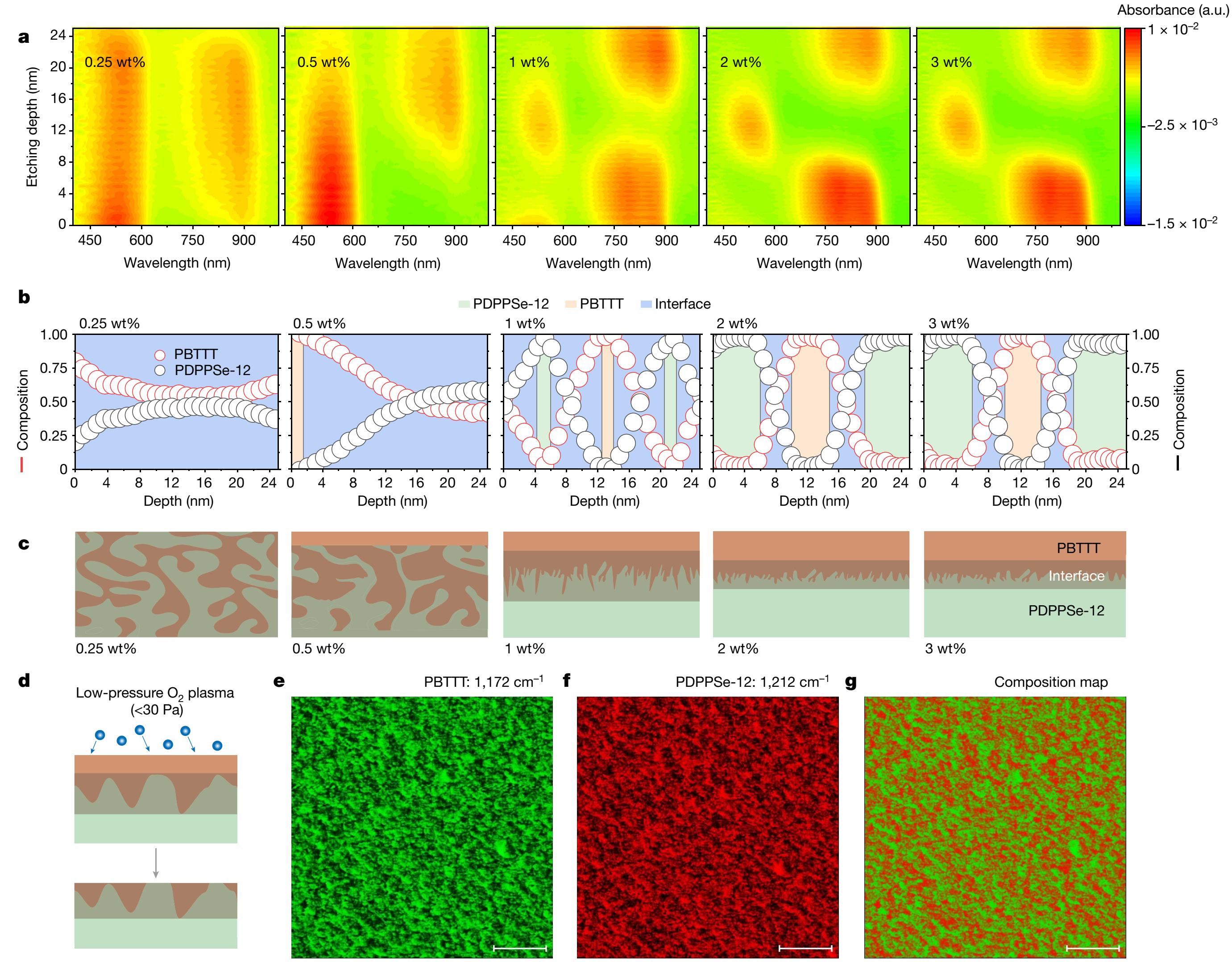

Fig. 2 | Reconstructed interface of PMHJ films. a, Optical intensity contours reconstructed from absorption spectra of PMHJ films with various 4Bx contents. The colours represent the absorbance during etching. The depth-dependent intensity contours were obtained by calibrating the etching time with the total film thickness. **b**, Variation of the two-polymer composition at different etching depths and contents of 4Bx. Here yellow, green and blue rectangles correspond to the PBTTT, PDPPSe-12 and the interface layers, respectively. The ultraviolet–visible absorption spectra reveal characteristic peaks of PDPPSe-12 and PBTTT at 870 nm and 540 nm, respectively. To determine the percentage content of PDPPSe-12 and PBTTT at different depths, we utilized compositional deconvolution and normalization methods. **c**, Interface schematic of PMHJ films at various crosslinker contents. As the content of 4Bx increased to 2 wt%, the thickness of the interface decreased to 4 nm, and the thicknesses of PDPPSe-12 and PBTTT on both sides was about 6 nm and about 4 nm, respectively, which is comparable to the calculated thicknesses from the TOF-SIMS measurements. **d**, Schematic of the oxygen plasma etching process to expose the PDPPSe-12 and PBTTT interface in a three-layer (6, 4, 4) PMHJ film (with layers arranged as follows from top to bottom: PBTTT layer, interface layer and PDPPSe-12 layer). **e**–**g**, PiFM images of the PBTTT (**e**) and PDPPSe-12 (**f**) domains of the interface layer. The green colour represents the PBTTT phase probed by the infrared laser at 1,172 cm^{-1} and the red colour represents the PDPPSe-12 phase probed at 1,212 cm^{-1}. **g**, PiFM overlay image of **e** and **f** to provide a chemical map of PDPPSe-12 and PBTTT at the interface layer. Scale bars, 300 nm.

their blending interface layers. Although the PMHJ shares a similar periodical geometry to the superlattice, the presence of propagons in less-ordered polymers and the rough interfaces produce significant interface scattering even in the in-plane direction (Fig. 1a and Supplementary Discussion II). Therefore, the in-plane κ_L can be effectively reduced by the size-effect-enhanced interface scattering when the individual layer's thickness is comparable to or ideally smaller than their phonon-like mean free path along the conjugated backbone and π–π stacking directions[28,29].

To implement this PMHJ concept, we utilized selenium-substituted diketopyrrolopyrrole (PDPPSe-12) and poly(2,5-bis(3-tetradecylthiophen-2-yl)thieno[3,2-b]thiophene) (PBTTT) as individual polymers owing to their high thermoelectric properties (Supplementary Figs. 1 and 2). A four-armed azide-based crosslinker (4Bx) was blended to crosslink the film to inhibit the aforementioned solvent corrosion and manipulate interfacial properties during sequential assembly. The PMHJ was indexed as (L_{P1}, L_I, L_{P2}), where L_{P1}, L_I and L_{P2} are the thickness of the PDPPSe-12, interface and PBTTT layer, respectively. The thickness of crosslinked PDPPSe-12 and PBTTT was controlled to be 6 nm and 4 nm, respectively, which corresponds to 2–3 molecular layers. The thicknesses are determined by the phonon-like mean free path in in-plane direction from non-equilibrium molecular dynamics (NEMD) simulations[30] (Supplementary Figs. 3–6, Supplementary Tables 1 and 2, and Supplementary Note I) and the experimental limitation to spin-coated films (Supplementary Fig. 7). Moreover, the interpenetrating interface layer was designed to be 4 nm, corresponding to approximately two mixed molecular layers to form the (6, 4, 4) PMHJ structure.

To construct the PMHJ, PDPPSe-12 or PBTTT solution with varying concentrations of 4Bx was spin-cast to form a film, which was then subjected to a photocrosslinking process that was confirmed through Fourier-transform infrared spectroscopy[31,32] (Supplementary Fig. 8). Specifically, the characteristic vibration peak of azide groups at 2,129 cm^{-1} appeared for pristine films but disappeared after exposure to ultraviolet light for 20 min, indicating an efficient photochemical reaction of the crosslinker inside the PBTTT and PDPPSe-12 films. To evaluate the capability of the crosslinked film to prevent solvent corrosion, we investigated the film retention of PBTTT and PDPPSe-12 at various 4Bx concentrations. With 2 wt% 4Bx, we obtained preservation of 80% and 90% for the PBTTT and PDPPSe-12 films, respectively, without affecting the doping uniformity (Supplementary Figs. 9–11). Notably, even at a high 4Bx concentration of 5 wt%, the film retentions of PBTTT and PDPPSe-12 were 83% and 94%, respectively, indicating that the pre-deposited film was lightly back-infiltrated with the second layer, which resulted in the desired interface layer in the multiple sequence coating processes.

Time-of-flight secondary ion mass spectrometry (TOF-SIMS) was used to characterize the cross-section of the constructed PMHJ films (Fig. 1e–h). Se^-, C_3N^-, CNO^- and CN^- were the characteristic components of the PDPPSe-12 layer whereas the PBTTT layer contained a higher percentage of C_4SH^- and C_2SH^-. The PDPPSe-12 and PBTTT are represented by green and red in Fig. 1e–g, respectively. By combining all mass intervals of the film, we constructed alternating layers with PDPPSe-12, PBTTT and the interface (Fig. 1h). The periodic structure was also observed using energy-dispersive X-ray spectroscopy in scanning transmission electron microscopy (STEM-EDS; Supplementary Fig. 12), which can be utilized to calibrate individual layer thickness. Statistical analysis revealed the thickness of PDPPSe-12, PBTTT and the interface layer to be 6.3 ± 0.5 nm, 4.2 ± 0.4 nm and 3.9 ± 0.4 nm, respectively (Supplementary Fig. 13). All these results demonstrate the alternating assembly of PDPPSe-12/interface/PBTTT, eventually forming the designed PMHJ structure. In particular, the periodic number can be extended to 100, resulting in a maximum thickness over 1,800 nm to satisfy diverse thickness requirements (Supplementary Figs. 14 and 15).

Film-depth-dependent light absorption spectroscopy was used to probe the vertical features of PMHJ films with varying 4Bx contents[33–35]. As pristine PDPPSe-12 and PBTTT films have different absorption peaks at 870 nm and 540 nm, respectively (Supplementary Fig. 16a), we determined the time- and thickness-dependent film composition by quantifying the in situ evolution of the nominal absorption during soft plasma etching. Figure 2a shows the optical intensity contours at each depth simulated from absorption spectra (Supplementary Fig. 16b–f). The introduction of 0.25 wt% 4Bx resulted in the appearance of both PBTTT and PDPPSe-12 at each depth, indicating bulk-heterojunction characteristics (Fig. 2b,c). With an increase in the 4Bx amount to 0.5 wt%, a pure PBTTT layer formed at approximately 1 nm at the top part, with a mixed layer thickness of up to 24 nm. At 1 wt% 4Bx, two alternate absorption peaks were observed, indicating the formation of a periodic structure of individual polymers. The interfacial layer was estimated to be 8 nm, compared with PDPPSe-12 and PBTTT layer thicknesses of 2 nm each. With a further increase in the 4Bx amount to ≥2 wt%, the cross-section behaviour remained nearly constant with a clear sandwich structure of PDPPSe-12 (6 nm)/interface (4 nm)/PBTTT (4 nm), which was consistent with the results of TOF-SIMS characterization.

Interface characteristics sandwiched by two individual layers are crucial for PMHJ films to manipulate propagon scattering. However, their direct mapping is challenging because they are buried internally. To address this issue, we carried out photo-induced force microscopy (PiFM)[36,37] characterizations directly on the interface that was exposed by a soft etching process (Fig. 2d and Supplementary Fig. 17). The wavenumbers 1,172 cm^{-1} and 1,212 cm^{-1} were chosen as the Fourier-transform infrared characteristic peaks of PBTTT and PDPPSe-12, respectively (Supplementary Fig. 18). The topography pattern of the interface showed distinct and uniform phase separation with small fibrous aggregates (Supplementary Fig. 19). After combining the PBTTT and PDPPSe-12 distributions, the interface shows a characteristic bulk heterojunction (BHJ) feature with a gently interpenetrated network (Fig. 2e–g and Supplementary Fig. 20), which is similar to other BHJs in polymeric solar cells[38,39]. The film-depth-dependent light absorption spectroscopy and PiFM characterization results collectively demonstrate the presence of a periodically hierarchical heterojunction geometry with sandwiched BHJ-like interfaces in the PMHJ film. This unique structure can enhance the diffusive scattering of propagons at the interface compared with the typical inorganic superlattice with atomic-level flatness (Supplementary Fig. 21), which illustrates the advantage of the PMHJ geometry.

Thermoelectric performance

To evaluate the thermoelectric conversion capability, we doped the PMHJ films with iron chloride ($FeCl_3$). Ultraviolet–visible–near-infrared spectra of the films revealed two broad polaronic bands at 1,370 nm and 2,000 nm, indicating efficient doping of PDPPSe-12 and PBTTT, respectively (Supplementary Fig. 22). More importantly, TOF-SIMS etching analysis showed obvious thickness-dependent alternating peaks and valleys in the signals of the PDPPSe-12 and PBTTT layers (Extended Data Fig. 1). The extracted individual thickness was consistent with that of the pristine film, indicating a well-maintained PMHJ structure, even upon chemical doping. In particular, we observed a distinct dopant distribution in the PDPPSe-12 and PBTTT layers owing to the different host–dopant interactions. Even with the introduced dopants, the PMHJ films preserve ordered molecular packing, as evidenced by the multi-ordered diffractions in grazing-incidence wide-angle X-ray scattering characterization (Supplementary Figs. 23–25). Consequently, the PMHJ film shows an apparent Hall mobility of up to 1.9 $cm^2\,V^{-1}\,s^{-1}$ along the in-plane direction (Supplementary Fig. 26), comparable to the individual values obtained for PDPPSe-12 and PBTTT films that range from 1.7 $cm^2\,V^{-1}\,s^{-1}$ to 1.9 $cm^2\,V^{-1}\,s^{-1}$.

Nanoresolved 3ω (third harmonic frequency) scanning thermal microscopy[40,41] was utilized to evaluate the heat transfer (out-of-plane and in-plane directions) of the PMHJ film compared with individual films (Fig. 3a–c). In contrast to a low $V_{3\omega}$ (voltage changes at the third harmonic frequency) of about 1 mV on PBTTT and PDPPSe-12 films, a higher $V_{3\omega}$ of about 6 mV across the PMHJ film was observed, because of less heat flow from the tip into the PMHJ sample (Supplementary Fig. 27), which confirms the lower κ in the PMHJ film. For an inorganic superlattice, pronounced phonon scattering is primarily expected to occur in the vertical direction, resulting in a significant suppression of out-of-plane thermal conductivity ($\kappa_\perp$)[20,42,43]. In contrast, owing to a rough blending interface with less-ordered molecular packing that led to diffused propagon scattering, PMHJ films hold the potential to suppress $\kappa_\parallel$ (refs. 28,44). The NEMD simulation conducted on the PMHJ structure demonstrated that the presence of disordered molecular packing and the interface exert a notable inhibitory effect on $\kappa_\parallel$ (Supplementary Fig. 28). Therefore, we quantitatively evaluated $\kappa_\perp$ and $\kappa_\parallel$ of undoped samples at room temperature (Supplementary Figs. 29 and 30, and Supplementary Note II). The $\kappa_\perp$ of the PMHJ film was suppressed to 0.06 $W\,m^{-1}\,K^{-1}$, representing a 57% decrease compared with PDPPSe-12 and a 70% reduction compared with PBTTT films. Interestingly, the $\kappa_\parallel$ of the PMHJ film was determined to be 0.10 $W\,m^{-1}\,K^{-1}$, which is 55% and 76% lower than that of undoped PDPPSe-12 (0.22 $W\,m^{-1}\,K^{-1}$) and PBTTT (0.41 $W\,m^{-1}\,K^{-1}$) films (Supplementary Fig. 31a), respectively. It worth noting that the exceptionally low $\kappa_\parallel$ was further verified by the independent suspended microdevice characterization and commercial institution (Supplementary Figs. 33 and 34). More importantly, the PMHJ concept is widely applicable for

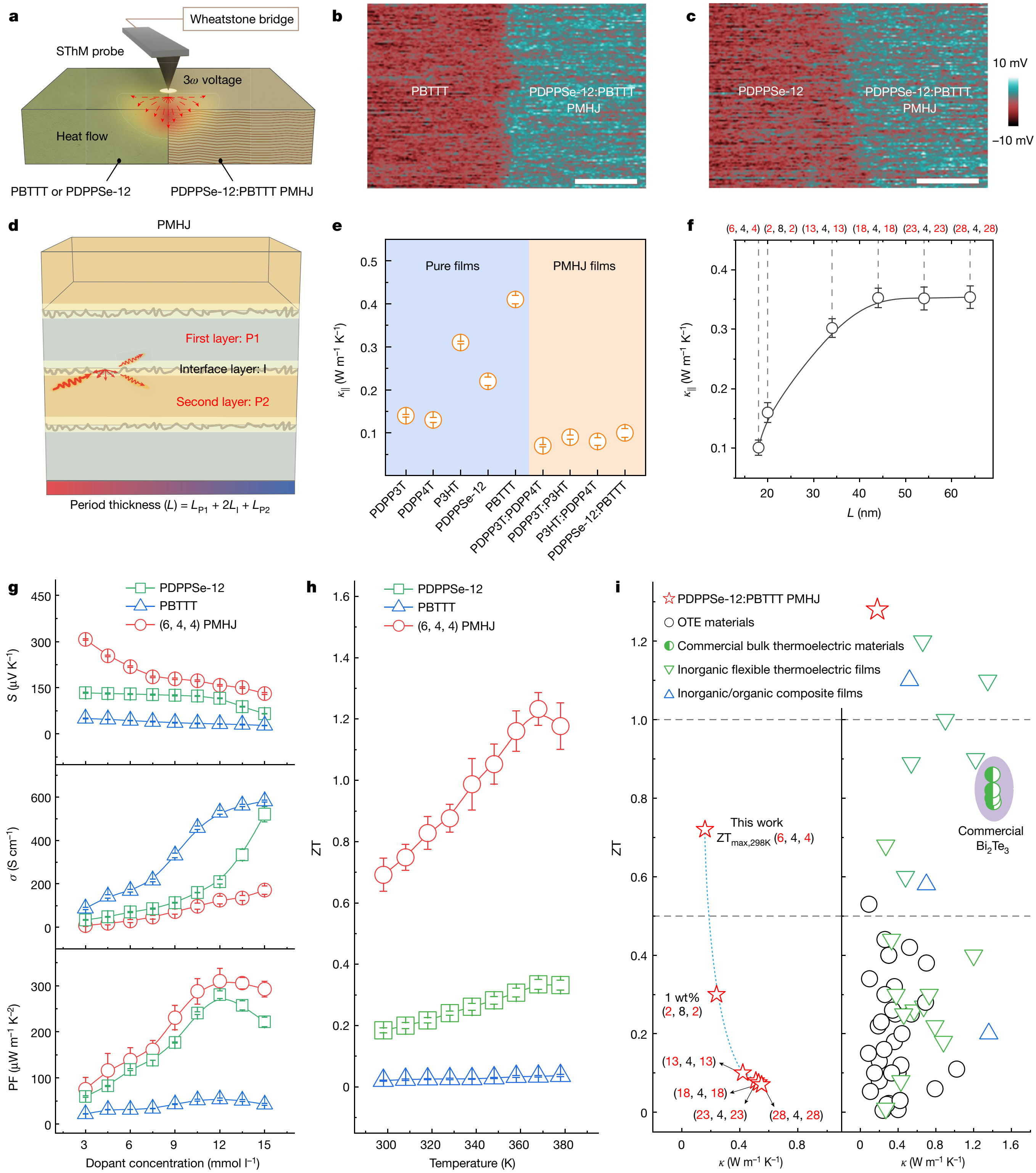

Fig. 3 | See next page for caption.

different combinations of polymers. For instance, in contrast to the $\kappa_{\parallel}$ of 0.14 W m^{-1} K^{-1}, 0.13 W m^{-1} K^{-1} and 0.31 W m^{-1} K^{-1} for PDPP3T, PDPP4T and P3HT, respectively, the $\kappa_{\parallel}$ of the combined (6, 4, 4) PMHJ was also reduced to approximately 0.10 W m^{-1} K^{-1} at room temperature (Fig. 3e).

Period thickness has profound effects on phonon scattering following the size effect in a typical superlattice[43,45,46]. To gain more insight, we investigated the relationship between the reduction of $\kappa_{\parallel}$ and the period thickness of the undoped PMHJ (Fig. 3d). Compared with the low $\kappa_{\parallel}$ of 0.10 W m^{-1} K^{-1} for the (6, 4, 4) PDPPSe-12:PBTTT PMHJ film, a significant increase in $\kappa_{\parallel}$ to 0.30 W m^{-1} K^{-1} occurred when the PMHJ was varied to (13, 4, 13). With a further increase in the individual thickness of the two polymers, the $\kappa_{\parallel}$ increased slightly to 0.35 W m^{-1} K^{-1} and eventually approached that of the bulk-film values (Fig. 3f). This phenomenon is consistent with the classical size effect[28,42], where the individual layer thickness approaches the mean free path of phonon vibrations, resulting in enhanced interface scattering and

Fig. 3 | Thermal transport properties and thermoelectric performance of PMHJ films. **a**, Schematic for scanning thermal microscopy (SThM) characterization. **b**,**c**, Nanoresolved 3ω scanning thermal microscopy thermal maps of PDPPSe-12:PBTTT (6, 4, 4) PMHJ and PBTTT (**b**) and PDPPSe-12:PBTTT (6, 4, 4) PMHJ and PDPPSe-12 (**c**) films. Scale bars, 2 μm. **d**, Schematic illustration of (L_{P1}, L_I, L_{P2}) PMHJ film. The single-period thickness L is determined by ($L_{P1} + 2L_I + L_{P2}$). The first layer, the interface layer, and the second layer are indicated in light green, light orange, and orange, respectively. **e**, $\kappa_{\parallel}$ of undoped PDPP3T, PDPP4T, P3HT, PDPPSe-12, PBTTT and their corresponding PMHJ films with a thickness of 540 nm. The $\kappa_{\parallel}$ was measured using a Linseis thin-film analyser set-up at room temperature. **f**, Period-thickness-dependent $\kappa_{\parallel}$ of undoped PDPPSe-12:PBTTT PMHJ film. **g**, Dopant-concentration-dependent S, σ and PF of $FeCl_3$-doped PDPPSe-12, PBTTT and 6-period PDPPSe-12:PBTTT (6, 4, 4) PMHJ films. The measurement was performed on a thermoelectric thin film with liquid nitrogen (TETF-LN) set-up (Supplementary Fig. 38). **h**, Temperature-dependent ZT of 12 mmol l^{-1} $FeCl_3$-doped PDPPSe-12, PBTTT and 6-period PDPPSe-12:PBTTT (6, 4, 4) PMHJ films. The films underwent annealing at 383 K for 3 min before measurements and S, σ and κ were characterized simultaneously on the same sample using the Linseis thin-film analyser system. **i**, ZT and κ of PDPPSe-12:PBTTT PMHJ films, typical organic thermoelectric (OTE) materials, flexible inorganic thermoelectric materials, inorganic/organic flexible composite films and commercial bulk thermoelectric materials at near room temperature. The dependence of the $\kappa_{\parallel}$ and ZT values on period thickness at room temperature is illustrated on the left side. The dopant concentration of all the PMHJ films depicted on the left side is 12 mmol l^{-1}. The right side includes comparisons of ZT and $\kappa_{\parallel}$ values for inorganic flexible materials, inorganic/organic flexible composite films and commercial bulk materials in the near-room-temperature range. The detailed data in the right portion of **i** and the relevant references are provided in Supplementary Table 3. The data in **e**–**h** are mean ± s.d. from three devices.

reduced κ. This size effect was also confirmed by NEMD simulations (Supplementary Fig. 35). Of particular note, the (2, 8, 2) film shows a slightly higher $\kappa_{\parallel}$ of 0.16 W m^{-1} K^{-1} compared with the (6, 4, 4) film. This discrepancy can also be attributed to its high-volume ratio of the blending layer, which has a higher κ value that is associated with the large domain size (Supplementary Figs. 31b and 32) and a relatively lower interface density of 0.15 nm^{-1} (that is, fewer interfaces per unit thickness).

With reduced lattice contribution to $\kappa_{\parallel}$, we anticipated a significant reduction in the total thermal conductivity for the doped films. The 12 mmol l^{-1} $FeCl_3$-doped PBTTT and PDPPSe-12 films showed a room-temperature $\kappa_{\parallel}$ of 0.79 W m^{-1} K^{-1} and 0.45 W m^{-1} K^{-1}, respectively, at σ of 520 S cm^{-1} and 252 S cm^{-1} at room temperature (Supplementary Fig. 36). In contrast, the $\kappa_{\parallel}$ of the doped PMHJ film was determined to be 0.15 W m^{-1} K^{-1} at σ of 128 S cm^{-1}, representing a more than 60% decrease compared with the individual doped films. A similar trend was also observed for the doped poly(diketopyrrolopyrrole-terthiophene) (PDPP3T), poly(diketopyrrolopyrrole-quaterthiophene) (PDPP4T) and poly(3-hexylthiophene-2,5-diyl) (P3HT) films, providing further evidence for the critical role of the PMHJ geometry in suppressing thermal conductivity under doping conditions to enhance thermoelectric performances (Supplementary Fig. 37).

The dopant-concentration and temperature-dependent S and σ of trademarked fluoropolymer (CYTOP-M)-encapsulated PDPPSe-12, PBTTT and 6-period (6, 4, 4) PMHJ films were compared along the in-plane direction (Fig. 3g and Supplementary Figs. 38–41). The room-temperature S and σ of both individual polymer films and PMHJ films showed a similar dependence on dopant concentration. With an increase in dopant concentration from 3 mmol l^{-1} to 15 mmol l^{-1}, the S of the PMHJ film gradually decreased from >300 μV K^{-1} to 131 μV K^{-1}, whereas the σ increased from 8 S cm^{-1} to 170 S cm^{-1}. The peak power factor (PF = $S^2\sigma$) at 298 K realized 347 μW m^{-1} K^{-2} at a concentration of 12 mmol l^{-1} to produce a ZT of 0.74. In particular, this PF surpassed the optimized PF value of 280 μW m^{-1} K^{-2} for pure PDPPSe-12 films. The temperature-dependent σ, S and $\kappa_{\parallel}$ for the (6, 4, 4) PMHJ films were evaluated simultaneously in a single device. The S showed a steady increase from 165 μV K^{-1} to 179 μV K^{-1}, which could be attributed to the thermally assisted activation mechanism in doped polymers. Simultaneously, σ increased by 52% to a maximum value of 196 S cm^{-1}, resulting in a maximum PF of 628 μW m^{-1} K^{-2} at 368 K. With a slight increase of $\kappa_{\parallel}$ to 0.18 W m^{-1} K^{-1}, the PMHJ films exhibited a maximum ZT of 1.28 at 368 K (Fig. 3h). This value is comparable to that of a thicker PMHJ film (ZT_{max} = 1.24 for 100-period (6, 4, 4) with a thickness of 1.8 μm; Supplementary Fig. 42). This performance not only promises polymeric and flexible thermoelectrics by breaking the milestone of ZT > 1 but also, more importantly, exceeds commercial thermoelectric materials in the near-room-temperature regime (Fig. 3i).

Compared with PDPPSe-12 and PBTTT films, the PMHJ films showed a much-reduced $\kappa_{\parallel}$ and a higher PF, mainly associated with the periodic structure. On the one hand, the interpenetrated interfaces induce size-effect-enhanced interfacial propagons scattering and significantly suppress κ_L, thus contributing dominantly to the ZT leap. On the other hand, the PMHJ might enable quasi-two-dimensional charge transport within two to three polycrystalline molecular layers. This is partly validated by the high overall Hall mobility of the PMHJ film, which is comparable to that of PDPPSe-12 and higher than that of PBTTT, even though the interfacial blending layers should have a much lower mobility. More interestingly, the electron-donating and electron-withdrawing nature of PBTTT and the DPP unit in PDPPSe-12, respectively, allow intermolecular charge transfer at their blending interface (Supplementary Fig. 43), thus bringing on a mixed density of states in the PMHJ compared with individual polymers to enhance electronic entropy. This, combined with a low carrier concentration (Supplementary Figs. 44 and 45), contributes to an enhanced S of PMHJ. Therefore, the significant suppression of $\kappa_{\parallel}$, along with the maintained high mobility and increased S, co-produce a substantial improvement in ZT values. Notably, a similar phenomenon was observed in the n-type PBFDO (poly(benzodifurandione)):n-PT3 (n-type polythiophene derivative)[47] PMHJ film (Supplementary Fig. 46), which illustrates that the PMHJ offers a general geometry to enhance thermoelectric performance for plastic thermoelectrics.

Flexible TEGs

In terms of application capability, the PMHJ films are bottlenecked by the structural complexity towards large-area fabrication. Here we demonstrate PMHJ devices fabricated by a multiple-step bar coating technique (Supplementary Fig. 47a). Each layer underwent a meniscus-guided coating and a photocrosslinking process. Six-period PMHJ films (3 × 3 cm^2) on a 2-inch silicon wafer showed excellent uniformity (Supplementary Fig. 47b), with excellent uniformity in total thickness (3% variation), individual layer thickness and chemical species (Supplementary Figs. 48–50). With these merits, we fabricated 3 × 10 cm^2 flexible 9-leg PMHJ devices on a polyimide substrate (Supplementary Fig. 51). The average S and σ were extracted to be 159 μV K^{-1} and 121 S cm^{-1}, respectively. When combined with the infrared-imaging-estimated κ of 0.15 W m^{-1} K^{-1}, the room-temperature ZT still reaches 0.61 ± 0.04 (Supplementary Fig. 52 and Supplementary Table 4). This corresponds to 88% of the optimized performance achieved by small-area films. More importantly, these devices could be easily folded into a corrugated shape. When worn on the human wrist (Fig. 4a,b), the device produced an open-circuit voltage of 3.3 mV. We also measured the power-generation capability in air (Fig. 4c, and Supplementary Figs. 53 and 54). The generators showed a maximum power output of 6 nW at T_{hot} (heat source temperature) = 36 °C (ΔT = 4.5 K), and the maximum power output increased with T_{hot}, reaching 522 nW at T_{hot} = 150 °C (ΔT = 38 K). The p-type-only TEG also marks one of the

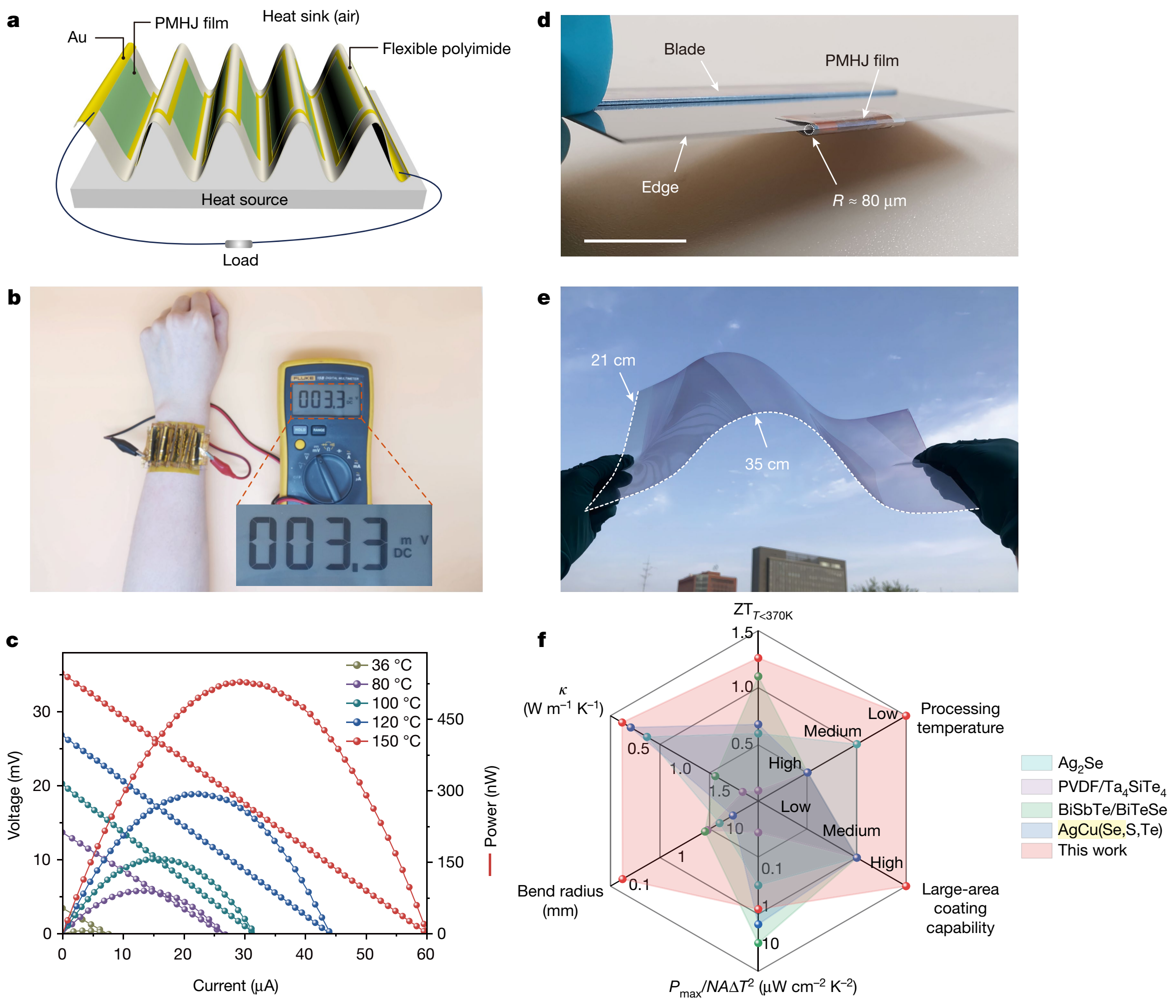

Fig. 4 | Solution-coated large-area PMHJ films and flexible generators. **a**, Schematic illustration of the flexible integrated all-polymer TEG and power-output measurement in air. **b**, Photograph of the 9-leg flexible thermoelectric device adhered on a human's wrist. **c**, The measured output voltage (straight line) and power (parabola line) of the integrated PMHJ TEG at different temperature differences. **d**, Photograph of a PMHJ film rolled on a commercially available blade with a bending radius of 80 µm. Scale bar, 1 cm. The thicknesses of the polyimide substrate and PMHJ film were 25 µm and 240 nm, respectively. **e**, Photograph of a large-area 1-period PMHJ film (35×21 cm^2) fabricated using the bar coating method. **f**, Radar charts for comparing the performance of typical flexible thermoelectric films. Evaluating indexes include $ZT_{T<370K}$, processing temperature, large-area coating capability, normalized power density ($P_{max}/NA\Delta T^2$, where N is the number of thermoelectric modules and A is the cross-sectional area), bend radius and κ. The chart includes inorganic flexible films (BiSbTe/BiTeSe, AgCu(Se,S,Te), Ag_2Se) and inorganic/organic composite films (PVDF (poly(vinylidene fluoride))/Ta_4SiTe_4). The PDPPSe-12:PBTTT PMHJ film is highlighted in red. The values are summarized in Supplementary Table 5.

highest normalized power densities (1.12 µW cm^{-2} K^{-2}) among TEGs based on organic materials or carbon nanotubes, even comparable to inorganic flexible TEGs[48–50].

With inherent flexibility, our PMHJ films on a 25-µm polyimide substrate show excellent mechanical resilience when subjected to twisting, bending and even folding (with a radius of curvature of 80 µm; Fig. 4d and Supplementary Fig. 55). It is worth noting that the σ of the PMHJ films remains nearly unchanged compared with the σ_0 even after 2,000 bending cycles at bending radii of 2 mm and 0.08 mm (Supplementary Fig. 56), where previously reported materials could only endure similar treatment for up to 2,000 cycles. More surprisingly, σ remains at 95% of its initial value even after 100,000 cycles, significantly surpassing previous striking results. In addition, the single-period PMHJ films can be coated over large areas. For instance, a 35×21 cm^2 coated film on a polyethylene terephthalate substrate, achieved through a simple bar coating method, shows a uniform thickness of 16.2 ± 2.6 nm (Fig. 4e and Supplementary Fig. 57) across different regions. These features, combined with low κ, high ZT, prominent power output and processing temperature below 400 K, position PMHJ film as an outstanding candidate for flexible thermoelectric materials (Fig. 4f and Supplementary Table 5).

In conclusion, our study demonstrates the robust effectiveness of the PMHJ concept in developing exceptional high-ZT plastic thermoelectric materials. By creating alternate sub-10-nm PDPPSe-12:PBTTT films with gently interpenetrated interfaces, the PMHJ films show prominent charge transport mobility, superior S and much enhanced propagon scattering at the interfaces. The PMHJ film showed a low $\kappa_{\parallel}$ of 0.18 W m^{-1} K^{-1} and a high PF of 628 µW m^{-1} K^{-2}, which combine to produce a maximum ZT of 1.28 at 368 K. The thermoelectric performance for the PMHJ film is superior to that of commercially available bulk materials and flexible thermoelectric candidates under the same temperature conditions. This result establishes a conceptual belief that nanostructured κ_L suppression is critical in overcoming the low-ZT limitation, even for weakly interacting plastics. More importantly, the PMHJ structure is also compatible with scalable coating techniques, paving an avenue to unlock ultraflexible plastic materials towards state-of-the-art and cost-effective wearable thermoelectrics.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-024-07724-2.

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Methods

Materials

PDPPSe-12 was synthesized by Z.L.'s group and D.Z.'s group. The molecular weight of the PDPPSe-12 polymer was measured to be 44.8 kDa, with a PDI (polydispersity index) of 3.1. 4Bx was provided by B.K.'s group. PBTTT was provided by I.M.'s group. The molecular weight of the PBTTT polymer was measured to be 30.1 kDa, with a PDI of 2.1. PBFDO was purchased from Volt-amp Optoelectronics Tech. and n-PT3 was provided by J.L.'s group.

Device fabrication

PDPPSe-12 and PBTTT were dissolved in *o*-dichlorobenzene at 110 °C at a concentration of 2 mg ml^{-1} and 6 mg ml^{-1}, respectively. A blend solution of PDPPSe-12 and 4Bx (2 wt%) was spin-coated on glass at 1,500 rpm for 1 min. After ultraviolet-light exposure (254 nm, 20 min) inside a nitrogen-filled glovebox, the blend solution of PBTTT and 4Bx (2 wt%) was spin-coated at 2,000 rpm for 1 min. After the same crosslinking and washing process, one-period PMHJ film was obtained. Repeating the above steps yielded layer-by-layer films with different periods. Subsequently, the layer-by-layer films were annealed at 180 °C for 1 h. To dope the films, we immersed them in $FeCl_3/CH_3NO_2$ with various concentrations for 30 s and then dried them under a stream of nitrogen. After doping, the samples were encapsulated with CYTOP-M (3:1) with a thickness of 500 nm. Subsequently, the samples were placed in a vacuum chamber for measurements.

STEM-EDS characterization

STEM-EDS specimens were prepared using a focused ion beam scanning electron microscope equipped with a cryogenic system (Helios NanoLabG3 CX, ThermoFisher Scientific). A representative area was selected and an initial protective layer of platinum was deposited on top of the region. A lamella was subsequently lifted to a focused ion beam transmission electron microscope half-grid and milled to a thickness of less than 100 nm at about −150 °C. STEM-EDS analysis was performed using a caesium-corrected transmission electron microscope operated at 300 kV (Spectra 300, ThermoFisher Scientific). The spatial distribution of selenium obtained from the EDS mapping carried information about the location of PDPPSe-12.

Film-depth-dependent light absorption spectroscopy characterization

The soft plasma etching procedure for depth-dependent ultraviolet and visible absorption spectroscopy characterization was carried out with argon gas at a pressure of <20 Pa by a Diener Femto plasma cleaner. The etching rate was correlated with pressure; in this work, the pressure was controlled at 20 Pa and the etching rate was about 10 nm min^{-1}.

TOF-SIMS characterization

TOF-SIMS analysis was conducted on a ToF-SIMS M6 mass spectrometer (ION-TOF). Mass spectra were collected by a 30 keV Bi_3^+ liquid metal cluster ion gun as an analysis beam in high-current bunched mode. The Bi_3^+ primary ion source was operated using a pulsed beam at 10 kHz (mass range 0–564 Da). Positive-ion spectra were calibrated by CH_3^+, $C_2H_3^+$, $C_2H_5^+$, $C_3H_5^+$ and $C_3H_7^+$, and negative-ion spectra were calibrated by O^-, OH^-, C_2^-, C_2H^- and Cl^-. An argon gas cluster ion gun (2.5 kV 2 nA) was applied when depth profile was recorded. Data were processed by ION-TOF SurfaceLab software (Version 7.3, ION-TOF) and the signal of corresponding ions or fragment ions was extracted to produce depth-dependent intensity profiles and three-dimensional or cross-sectional images. The signals were displayed on a colour scale, which were directly related to the intensities of detected ions of interest.

Ultraviolet–visible–near-infrared and Fourier-transform infrared characterization

The 350–2,000 nm ultraviolet–visible–near-infrared spectra were measured with a Hitachi UV-Visible/NIR Spectrophotometer UH4150 at room temperature. The baseline was collected with a blank substrate of quartz glass. The Fourier-transform infrared samples were deposited on KBr tablets and then measured in transmission mode with a VERTEX 70 v Fourier-transform infrared spectrometer with a resolution of 2 cm^{-1}.

Ultraviolet photoelectron spectroscopy measurements

The ultraviolet photoelectron spectroscopy measurements were performed on an Axis Ultra DLD (Kratos) spectrometer with an unfiltered He I (21.22 eV) excitation source and a pass energy of 5 eV. A bias voltage of −9 V was applied to the sample for obtaining the secondary electrons cut-off region.

Photo-induced force microscopy measurements

PiFM measurements were conducted on a Bruker Anasys nanoIR3 Instrument in the tapping mode. Pulsed infrared irradiation at a fixed wavelength was used to examine the sample, with the probe collecting infrared absorption data at that wavelength to generate infrared absorption images. This method allows for the mapping of chemical group distributions corresponding to that wavelength on the sample surface. Specifically, the atomic force microscope probe is positioned over a certain location, and the infrared laser sweeps through a range of infrared frequencies. The scan resolution was set to 512 × 512 pixels with a scan rate of 0.5 Hz. The data were evaluated and depicted with Gwyddion 2.61 scanning probe microscopy software.

Transient infrared measurements

The infrared images were measured with suspended device. First, a layer of poly(sodium 4-styrenesulfonate) (PSSNa) was spin-coated on a hydrophilic substrate. Then an organic semiconductor layer was prepared on it. It was then immersed in deionized water, and the PSSNa layer dissolved in the water. The floating organic semiconductor layer was transferred to a hydrophilic suspended substrate.

The infrared images of the suspended films were measured at room temperature and under a high-vacuum environment (5×10^{-4} Pa). A direct-current heating signal was applied to the device to create a heat source and form heat transfer in the film. A trigger signal was simultaneously applied to infrared camera to ensure temporal synchronization. The infrared image was recorded at a frequency of 1,000 frames per second.

Thermoelectric property measurements

The thermoelectric thin film with liquid nitrogen (TETF-LN) set-up was utilized for conducting thermoelectric property measurements. For electrical conductivity and Seebeck coefficient measurements, the channel length and width of the gold electrodes were 500 µm and 5,000 µm, respectively. A four-probe method was performed to determine the σ with an Agilent B2902A and Keithley 2182A. The S can be calculated by $S = \Delta V/\Delta T$, where ΔV is the thermal voltage between the two electrodes of the device when applied a temperature difference of ΔT. The ΔV is measured by a Keithley 2182A in vacuum. The temperature gradient was introduced by Peltier elements. An infrared camera (temperature sensitivity <50 mK, FLIR A300) was used to detect the temperature difference between two electrodes (channel length, 500 µm). Furthermore, the temperature difference was corrected as follows. First, we measured the temperature of a glass substrate at different temperatures using both FLIR A300 and commercial Pt 100 simultaneously and the calibration curve was built up (Supplementary Fig. 39a). Thereafter, the temperature difference between the two electrodes, which was proportional to the current applied to the Peltier elements, was obtained by a calibrated infrared camera (Supplementary Fig. 39b).

The $\kappa_{\parallel}$ was measured using a Linseis thin-film analyser system (Supplementary Note II). Of particular note, the optimal performance was obtained through determining the temperature-dependent σ, S and $\kappa_{\parallel}$ on the Linseis thin-film analyser system using the same sample. The PMHJ films were fabricated on a Van der Pauw chip. To improve stability, a 500-nm CYTOP-M encapsulation layer was applied. The impact of CYTOP-M on determining the $\kappa_{\parallel}$ for the encapsulated samples was carefully considered. Separate tests were carried out to measure the thermal conductivity of both the CYTOP-M and the PMHJ (CYTOP-M encapsulated) films under consistent testing conditions and film thickness for CYTOP-M. By factoring in the thermal conductivity contribution of CYTOP-M, the data obtained from CYTOP-M served as a reference baseline during the data analysis phase. This approach effectively eliminated the influence of CYTOP-M on the final test results.

Data availability

All data supporting the findings of this study are available in the paper and its Supplementary Information. Source data are provided with this paper. Other related raw data are fully and freely available from the corresponding authors at the point of publication.

Code availability

All codes supporting the findings of this study are available in the Supplementary Information. Other related codes are fully and freely available from the corresponding authors at the point of publication.

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Author contributions C.-a.D. conceived the idea and designed the project. D.Z. supervised the project. L.-D.Z. commented on the experimental result, designed the integrated devices and analysed the mechanism. Dongyang Wang and J.D. prepared the samples and measured the thermoelectric performance, Hall mobility, film-depth-dependent light absorption spectroscopy and thermal conductivity. Dongyang Wang fabricated the large-area TEG and measured the output performance. Dongyang Wang, J.D., X.D., X.Z. and Yue Zhao performed theoretical calculations. Dong Wang and C.X. conducted the NEMD simulations and contributed to the data interpretation. Y.D. carried out the investigation on the iron chloride doping stability and fabricated PBFDO:n-PT3 PMHJ films. Yao Zhao carried out the TOF-SIMS characterizations. X.D. and Ye Zou carried out UPS measurements. Dongyang Wang and L.L. carried out SThM measurements. X.Y., Zhiyi Li and Yue Zhao performed the in-plane thermal conductivity measurements using the suspended microelectrodes. C.-a.D., Dongyang Wang, J.D., Y.M. and L.D.Z. analysed all the experimental data and wrote the paper. Dongyang Wang and F.Z. carried out the XRR measurements. C.-a.D., Dongyang Wang, Y.M., L.D.Z. and C.C. commented on the concept of phonons and the heat transport mechanism. Zitong Liu and D.Z. synthesized PDPPSe-12 and provided the suggestion to optimize the material structure. I.M. provided PBTTT and commented on the mechanism. M.L. and B.K. synthesized 4Bx, suggested the mechanism and revised the paper. All authors discussed and revised the paper.

Competing interests Dongyang Wang and C.-a.D. have filed a patent application on this work (202410500369.2). The other authors declare no competing interests.

Additional information

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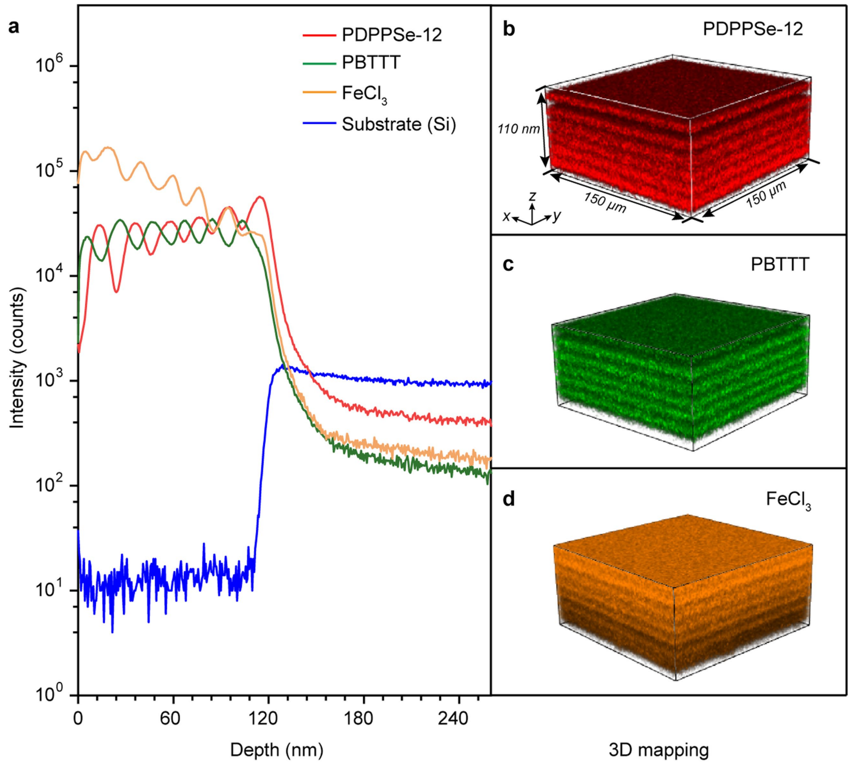

Extended Data Fig. 1 | TOF-SIMS analysis of doped (6,4,4) PMHJ films. **(a)** The depth profiles of the ion fragments of 12 mmol/L $FeCl_3$ doped PMHJ films. **(b-d)** TOF-SIMS 3D spatial configuration of (b) PDPPSe-12, (c) PBTTT, and (d) $FeCl_3$ signals constructed from (a) the depth profiles of PMHJ films. The red, green and orange colours represent the distribution of PDPPSe-12, PBTTT, and $FeCl_3$, respectively.